

LVA-008 — root-cause analysis

This is the in-depth analysis backing the filing-ready report in `README.md` and the standalone sample in `MinimalRepro.kt`. Every claim is anchored to in-repo device evidence; nothing here is surmised (§11.4.6). Where a thing is not confirmed it is marked `UNCONFIRMED`: explicitly.

1. What the bug is

`androidx.navigation:navigation-compose` crashes the process at **Activity destroy** when a **nested NavHost** (mounted from a parent `NavBackStackEntry`, the bottom-nav pattern) has a child entry that is still `INITIALIZED`:

```
java.lang.IllegalStateException: State must be at least 'CREATED'
to be moved to 'DESTROYED',
    but was 'INITIALIZED' in component
    NavBackStackEntry(route=search/search_input?query={query}
&categories={categories}
    &author_name={author_name}&author_id={author_id}
&sort={sort}&order={order}&period={period})
    at
androidx.lifecycle.LifecycleRegistry.checkLifecycleStateTransition()
    at androidx.navigation.NavBackStackEntry... updateState
    at androidx.navigation.internal.NavControllerImpl ...
    lifecycleObserver // host ON_DESTROY observer
```

Rethrown as `RuntimeException`: Unable to destroy activity `MainActivity` inside `ActivityThread.performDestroyActivity`.

2. The lifecycle-ordering root cause

```
NavBackStackEntry.effectiveState = min(hostLifecycleState,
maxLifecycle), ordinals
DESTROYED(0) < INITIALIZED(1) < CREATED(2) < STARTED(3) <
RESUMED(4).
```

1. The outer `NavHost` is on a graph destination (a bottom-nav tab). That parent entry reaches `RESUMED` and composes the **inner NavHost** (`start = list`).

2. The user navigates the inner host list -> search_input. The inner controller now tracks list (stopped) plus search_input.
3. Because of how the nested controller restores/transitions entries, a search_input entry can be **tracked while still INITIALIZED** — queued/restored but never driven up to CREATED (the not-yet-promoted / restored / transitioning copy that survives after the user's composed instance was popped).
4. The **Activity is destroyed**. The parent NavBackStackEntry (the inner host's LifecycleOwner) collapses **directly to DESTROYED** — it does not walk its nested children RESUMED->STARTED->CREATED->DESTROYED.
5. The inner controller's host observer calls `updateState()` / `maxLifecycle = DESTROYED` on the INITIALIZED search_input entry. `LifecycleRegistry.checkLifecycleStateTransition` rejects it: **there is no lifecycle event that moves an owner down from INITIALIZED** — `ON_DESTROY` requires at least `CREATED`. It throws.

This is the INITIALIZED-has-no-downward-event invariant (umbrella b/244910446, sibling b/178029606), surfaced via the **nested-host + Activity-destroy** path.

3. Why it surfaces in tests but is a real bug (not a test artifact)

Instrumentation (ActivityScenario / connectedAndroidTest) does a rapid create -> drive -> **destroy** of MainActivity, which reliably destroys the Activity while the inner search_input entry is still INITIALIZED. That is the *trigger*, not the *cause*:

- The stranded entry persists INITIALIZED for the **whole session** after the composed instance is popped — confirmed by the candidate-2 experiment (section 4 item 4).
- The same destroy path is crossed by real users on rotation / dark-mode / locale change / "Don't keep activities" / process death after using search.
- **Crashlytics confirms it live in the field**: FATAL, 4 events / 4 users on 1.3.11-1075 (per docs/CONTINUATION.md).
- It reproduces on clean cold-boot device runs, version-independently.

So the test environment makes a genuine lifecycle-ordering defect deterministic.

4. Falsified-fix ledger (8 device-verified dead ends)

All speculative changes were REVERTED per section 6.T.1. Evidence under `.lava-ci-evidence/genymotion/` and the per-candidate gate dirs.

#	Hypothesis	Result	Evidence
1	Bump nav 2.9.1 -> 2.9.8	FALSIFIED — identical ISE	<code>.lava-ci-evidence/genymotion/c11nav298-r2-20260608T16</code>
2	Bump nav 2.10.0-alpha04/05	FALSIFIED — identical ISE	release-notes di fix listed)
3	LenientTeardownRule / try-catch (walk cause chain)	Structurally impossible — rethrown as <code>RuntimeException("Unable to destroy activity")</code> in <code>Instrumentation.callActivityOnDestroy -> ActivityThread.performDestroyActivity;</code> process death, never propagates synchronously	<code>.lava-ci-evidence/genymotion/c11fix-20260608T1</code>
4	Move route to the OUTER NavHost	FALSIFIED — crash moved to the outer <code>search_input</code> (route <code>search_input</code> , no <code>search/</code> prefix) but still fired -> disproves "nesting strictly required"; bug is intrinsic to an INITIALIZED entry	<code>.lava-ci-evidence/genymotion/lvacand2-*</code>
5	Atomic-replace nav (<code>popUpTo(current) {inclusive=true} + navigate()</code>)	FALSIFIED — entry still tracked INITIALIZED at host-destroy	<code>.lava-ci-evidence/genymotion/c11atomicfix-r2-20260608T17</code>
6	NavTeardownGuard: on ON_STOP force-advance INITIALIZED entries in <code>currentBackStack.value</code> to CREATED	FALSIFIED — the phantom entry is not in the public <code>currentBackStack StateFlow</code> ; it lives in <code>@RestrictTo</code> internals (<code>back-queue / transitions-in-progress / saved-state-restore</code>), reflection-only, not shipped	<code>.lava-ci-evidence/genymotion/lvateardownguard-</code>
7	Activity-scoped inner <code>LocalLifecycleOwner</code> (keep <code>ViewModelStore / SavedStateRegistry</code> on parent entry)	FALSIFIED on the section 6.Z gate 2026-06-25 — C06 + C11 both reproduce 3x ISE on build 1074 @ 1310a922; reverted	<code>.lava-ci-evidence/1074-gate/{GATSUMMARY.json,C</code>
8	<code>launchSingleTop</code> dedupe on the <code>search_input</code> navigation	FALSIFIED — C06 + C11 FAIL	<code>.lava-ci-evidence/lva008-cand8-g{c06,c11}/</code>

Latest re-confirmation: `.lava-ci-evidence/1076-repro/` — Challenge52 (search chips) and Challenge48 (sync toggle) both crash with the identical ISE, showing the defect is **systemic across nested routes** (search/search_input and provider_config), firing at instrumentation activity-destroy.

5. Why each class of "fix" fails (the structural reasons)

- **AndroidX version bump fails** because no released navigation-compose (through 2.9.8 / 2.10.0-alpha05) changes the INITIALIZED → DESTROYED rejection in the nested host-observer walk. The fix has to land upstream.
- **onException / LenientTeardownRule / any TestRule fails** because the ISE is raised on the **main-thread Looper** inside `ActivityThread.performDestroyActivity` and rethrown as `RuntimeException("Unable to destroy activity")`. It is never an exception the JUnit statement chain or a `Thread.UncaughtExceptionHandler` can intercept synchronously — the instrumentation process simply dies ("Instrumentation run failed due to Process crashed"). It is, by construction, test-side uncatchable.
- **App-level NavHost guards fail** because the stranded entry is not reachable through any public API at the moment teardown runs (item 6 — not in currentBackStack; in @RestrictTo internals). A reflection-based prune would be fragile and break across nav releases, so it is not shipped.

6. Candidate real fixes

1. **AndroidX upstream fix (correct fix)**. `NavControllerImpl`'s host-lifecycle observer should no-op (or first drive up to CREATED) when asked to move an INITIALIZED nested entry to DESTROYED, instead of calling `LifecycleRegistry` with an illegal transition. File the report in `README.md` (+ attach `MinimalRepro.kt`); track against b/244910446.
2. **AndroidX Test Orchestrator process-isolation (test-side mitigation only)**. Running each test in its own process would contain the process-death so one teardown crash does not poison the run — but it does **not fix the user-facing crash**, and the Orchestrator/services APKs are **NOT cached offline** on the gate host, so it cannot be adopted under the Local-Only CI/CD constraint without a network fetch the operator must authorize. Mitigation, not a fix.
3. **Single-NavHost refactor (app-side, large blast radius)**. Collapse the nested search graph so `search_input` lives on one flat host, eliminating the parent-entry-as-LifecycleOwner pre-condition. This is the only app-

side avenue not yet fully shipped; the candidate-7 "single-NavHost collapse" partial was device-tested (.lava-ci-evidence/lva008-cand7-gate/) and C06/C11 still FAILED for that partial, so a *correct* full refactor would have to remove **every** nested host on the failing paths — high risk, operator-gated.

7. Current disposition

- Accepted as an **upstream androidx-navigation defect**; instrumented in production via NavTeardownCrashReporter (chained uncaught-exception handler in `LavaApplication.onCreate`) which tags the known ISE with section 6.AC Crashlytics context (feature=navigation, operation=activity-teardown, screen=search_input, lva_id=LVA-008) before the process dies — it instruments, never swallows.
- The upstream report (this directory) is ready to file; the operator files it (Google sign-in required).
- Workable item **LVA-008** remains OPEN until an upstream fix or the full single-NavHost refactor lands and C06 + C11 pass on the device gate.